

Float level transmitter For Liquid Application

Model: MFLT-1100L



PRODUCT DESCRIPTION

The Float Liquid Level Transmitter – MFLT-1100L is designed for continuous measurement of liquid level in tanks, vessels, and industrial process systems. The transmitter uses a float mechanism that moves with the liquid level and provides a corresponding electrical signal for level indication and monitoring. With 5 mm or 10 mm resolution, the MFLT-1100L provides dependable level measurement for a wide range of liquid applications. It is available with different measuring lengths, float materials, and process connections to suit specific application requirements.

The transmitter is suitable for water, oil, chemicals, and other compatible industrial liquids, making it a practical solution for continuous level monitoring in storage and process tanks.

KEY FEATURES

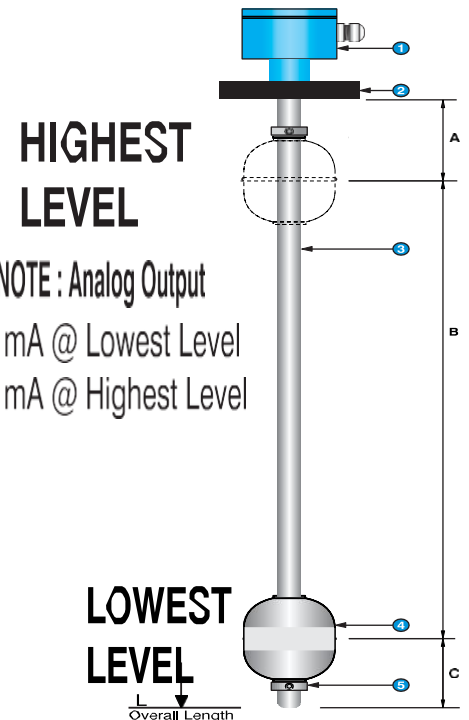
- Continuous liquid level measurement
- Float-operated sensing principle
- Reliable and accurate level measurement
- Resolution: 5 mm / 10 mm
- 4–20 mA output available
- Wide range of measuring lengths
- Suitable for various liquid applications
- Compact and robust construction
- Easy installation and maintenance
- Customized process connections available
- Suitable for storage and process tanks
- Float and wetted parts available in different materials

PROBE CONSTRUCTION

The Float Level Transmitter consists of a non-magnetic sealed stem containing a series of reed switches and resistors, a float with an integral magnet, a mounting adaptor or flange, and an enclosure containing the electronic circuit. As the liquid level changes, the magnetic float moves along the stem, activating the corresponding reed switches and generating a level-proportional electrical output.

OPERATING PRINCIPLE:

The MFLT-1100L Float Level Transmitter operates on the magnetic float and potentiometer principle. A float fitted with a permanent magnet moves vertically along a non-magnetic sealed stem in accordance with the liquid level. The movement of the float changes the position of the sensing element in the potentiometer, producing a variable resistance proportional to the liquid level. The electronic circuit converts this resistance signal into a continuous 4–20 mA output, providing reliable and continuous liquid level measurement.



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TYPICAL APPLICATIONS:

Our Float Level Transmitters are designed for continuous liquid level monitoring of water, diesel, lubricating oils, fuels, chemicals, and petrochemical liquids.

Here are just a few areas where the transmitter can be used:

- Water storage and treatment plants
- Diesel and fuel storage tanks
- Lubricating oil and hydraulic oil tanks
- Chemical storage tanks
- Petrochemical process tanks
- Industrial process vessels
- Effluent and wastewater treatment systems
- Cooling water and utility tanks
- Boiler and service water tanks
- General industrial liquid storage applications

LIMITATIONS:

- Not recommended for highly viscous or heavily sludge-laden liquids that may restrict float movement.
- Not suitable where excessive turbulence or strong agitation can affect float movement.
- Float and wetted materials must be compatible with the process liquid.
- Maximum operating temperature and pressure must be within the specified limits.
- Proper installation and alignment of the stem and float are required for reliable measurement.
- The measuring range is limited by the specified stem length.
- Magnetic or strong electromagnetic interference near the sensing stem should be avoided.
- Not recommended for applications where heavy deposits or solids can accumulate on the float or stem.

TECHNICAL DATASHEET

MFLT-1100L – Float Liquid Level Transmitter

Parameter	Specification
Instrument Type	Float Liquid Level Transmitter
Model	MFLT-1100L
Application	Continuous Liquid Level Measurement
Measuring Principle	Magnetic Float & Potentiometer
Probe length	300 mm to 5000 mm
Resolution	5 mm / 10 mm
Accuracy	1%
Float Movement	Vertical
Float Type	Magnetic Float
Stem Type	Non-Magnetic Sealed Stem
Stem Material	SS304 / SS316 / Other, as required
Stem Diameter	12.7 mm
Float Material	SS304 / SS316 / PP / Other, as required
Float Diameter	40 mm / As per Liquid Specific Gravity
Process Connection	Threaded / Flanged / Tri-Clamp / Other – As per Customer Requirement
Mounting	Top Mounting

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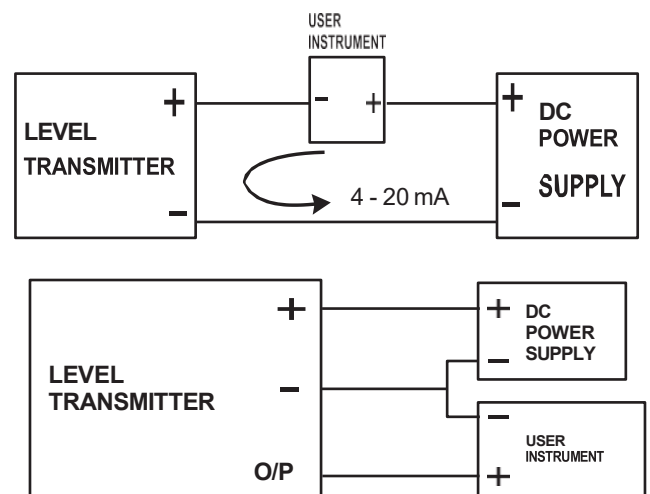
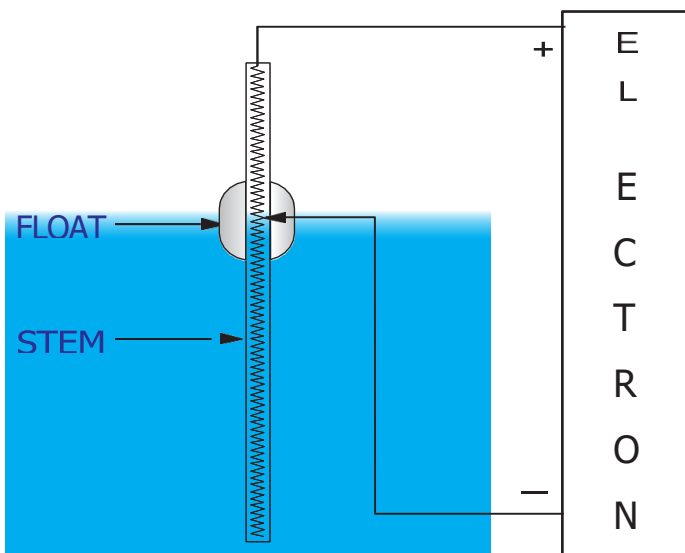
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Parameter	Specification
Output	2-Wire 4–20 mA / 0–10 VDC / RS-485 / Resistance – As per Customer Requirement
Power Supply	24 VDC, as applicable
Electrical Connection	Cable / Connector
RS-485 Communication	Available as per Customer Requirement
Load Resistance	As per Output Configuration
Enclosure	Weatherproof IP66 as per IS 2147; Flameproof, Gas Group IIA & IIB as per IS 2148
Enclosure Material	Aluminium / SS / Engineering Polymer, as applicable
Operating Temperature	150 Deg /As per Float and Process Material
Operating Pressure	10 kg / As per Process Connection and Construction
Liquid Specific Gravity	As per Float Selection
Wetted Parts	SS / PP / Other Compatible Materials
Suitable Liquids	Water, Diesel, Lubricating Oils, Fuels, Chemicals & Petrochemical Liquids
Installation	Vertical Top Entry
Indication	Optional Local Display
Calibration	Factory Calibrated
Zero / Span Adjustment	As Applicable to Output Configuration
Cable Entry	13.5 mm /1/2" NPT /3/4"ET/Cable Gland Connector
Maintenance	Low Maintenance
Customization	Measuring Length, Float, Connection, Output & Material

TYPICAL WIRING

A : 2 wire 4 - 20mA

B : 3 wire 0 - 10V DC



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ORDERING INFORMATION – MFLT-1100L

1. OUTPUT

Code	Output
1	2-Wire, 4–20 mA
R	Resistive Output
4	RS-485 Digital Output

2. RESOLUTION

Code	Resolution
5	5 mm
10	10 mm

3 FLOAT SELECTION MOUNTING

Code	Float Description
3	34 mm Nylon Foam Float
4	40 mm Dia. SS316 Float
5	52 mm Dia. SS316 Float
6	68 mm Dia. SS316 Float
P	50 mm Dia. PP Float
PP	– Polypropylene

4 PROCESS CONNECTION

Code	Mounting / Process Connection
10	100 mm Dia. × 80 mm PCD Flange
75	75 mm Dia. × 60 mm PCD Flange
1D	1" Table 'D'
1A	1" ANSI 150# Flange
2D	2" Table 'D'
2A	2" ANSI 150# Flange
ND	2" Table 'D' – Non-Standard
NA	2" ANSI 150# Flange – Non-Standard
4B	4" BSP
1B	1" BSP
NB	1" BSP – Non-Standard
2B	2" BSP
NS	Non-Standard Mounting

5. ENCLOSURES

Code	Enclosure Type
W	Weatherproof, IP66
F	Flameproof, Gas Group IIA & IIB

Standard: Weatherproof IP66 as per IS 2147;
Flameproof, Gas Group IIA & IIB as per IS 2148.

6. MEASURING LENGTH

Code	Description
L	Measuring Length – As per Customer Requirement

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ORDERING CODE

MFLT-1100L / Output / Resolution / Float / Mounting / Enclosure / Measuring Length

CODE FORMAT

MFLT-1100L / O / R / F / M / E / L

Where:

- O = Output Code
- R = Resolution Code
- F = Float Selection Code
- M = Mounting / Process Connection Code
- E = Enclosure Code
- L = Measuring Length

Example

Mflt-1100l / 1 / 5 / 4 / 1a / W / 3000 Mm

1 = 2-Wire, 4–20 Ma

5 = 5 Mm Resolution

4 = 40 Mm Dia. Ss316 Float

1a = 1" Ansi 150# Flange

W = Weatherproof Ip66

3000 Mm = Measuring Length

Note: Measuring Length, Resolution, Float Material/Size, Mounting Arrangement, Enclosure And Output Can Be Customised As Per Application Requirements.

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